Transportation Model

Introduction

Transportation costs are an important component of pricing decision of any product; thus, it affects both the movement of raw materials or finished goods. If a facility will be the sole source or destination of shipments, the company can include the transportation costs in a locational cost–volume analysis by incorporating the transportation cost per unit being transported into the variable cost per unit. (If raw materials are involved, the transportation cost must be converted into cost per unit of output in order to correspond to other variable costs.) When a problem involves shipment of goods from multiple sending points to multiple receiving points, and a new location (sending or receiving point) is to be added to the system, the company should undertake a separate analysis of transportation. In such instances the transportation model which is special type of linear programming model is very helpful. It is a special-purpose algorithm used to determine the minimum transportation cost that would result if a potential new location were to be added to an existing system. It also can be used if a number of new facilities are to be added or if an entire new system is being developed. The model is used to analyze each of the configurations considered, and it reveals the minimum costs each would provide. This information can then be included in the evaluation of location alternatives.

Nature of Transportation Model

A transportation model can either be **balance** or **unbalance**. Balance transportation model is a situation the total demand is equal to the total supply, which is the column total is equal to row total. A situation occurring quite frequently in real-life problems is the case in which total demand is not equal to total supply, such situation known as unbalance transportation model. These *unbalanced problems* can be handled easily by the Initial basic solution procedures if we first introduce dummy sources or dummy destinations. In the event that total supply is greater than total demand, a dummy destination (warehouse), with demand exactly equal to the surplus, is created. If total demand is greater than total supply, we introduce a dummy source (factory) with a supply equal to the excess of demand over supply. In either case, shipping cost coefficients of zero are assigned to each dummy location or route because no shipments will actually be made from a dummy factory or to a dummy warehouse. Any units assigned to a dummy destination represent excess capacity, and units assigned to a dummy source represent unmet demand.

Solution Procedure:

(a) **Initial Basic Feasible Solution**

(i) North-West Corner Method,

(ii) Least Cost Method, and,

(iii) Vogel approximation Method.

(b) **Optimum Solution**

(i) Stepping Stone Algorithm Method

(ii) Modified Distribution Method

Illustration: Olamilekan Nigeria Limited transports products from three (3) factories to 4 (four) warehouses of its customers, the supply capacity of its factories, cost of transportation as well as the demand of the 4 (four) warehouses are presented in the table below;

Origin/destination	**D_1**	**D_2**	**D_3**	**D_4**	**Supply**
S_1	190	300	500	100	**700**
S_2	700	300	400	600	**900**
S_3	400	80	700	200	**1800**
DEMAND	**500**	**800**	**700**	**1400**	

You are required to formulate the problem as:

(i) a linear programming model and

(ii) determine the initial basic feasible solution using:

Northwest Corner Method,

Least Cost Method, and,

Vogel approximation Method.

Solution:

Transportation problem as linear programming model for Badagry Nigeria Limited.

Minimize Z: $190X_{11} + 300X_{12} + 500X_{13} + 100X_{14} + 700X_{21} + 300X_{22} + 400X_{23} + 600X_{24} +$

$400X_{31} + 80X_{32} + 700X_{33} + 200X_{34}$ Objective Function

Subject to:

$X_{11} + X_{12} + X_{13} + X_{14} = 700$
$X_{21} + X_{22} + X_{23} + X_{24} = 900$ } Supply constraint
$X_{31} + X_{32} + X_{33} + X_{34} = 1800$

$X_{11} + X_{21} + X_{31} = 500$
$X_{12} + X_{22} + X_{32} = 800$
$X_{13} + X_{23} + X_{33} = 700$ } Demand Constraint
$X_{14} + X_{24} + X_{34} = 1400$

$Xij \geq 0$ ϒ ij Non-negativity constraint

(i) North-west corner Method

	D_1	**D_2**	**D_3**	**D_4**	**Supply**
S_1	190 a **500**	300 b **200**	500	100	**700**
S_2	700	300 c **600**	400 d **300**	600	**900**
S_3	400	80	700 e **400**	200 f **1400**	**1800**
DEMAND	**500**	**800**	**700**	**1400**	**3400**

Row + Column – 1 = Number of Allocation

Quantity allocated		Cost/Unit ₦	Total ₦
(a) 500	*	190	= 95, 000
(b) 200	*	300	= 60, 000
(c) 600	*	300	= 180, 000
(d) 400	*	300	= 20, 000
(e) 400	*	700	= 280, 000
(f) 1400	*	200	= 120, 000
Total			**1,015, 000**

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(ii) Least Cost Method

	D_1	**D_2**	**D_3**	**D_4**	**Supply**
S_1	190	300	500	100 b	**700**

				700	
S_2	700 f **200**	300	400 d **700**	600	**900**
S_3	400 e **300**	80 a **800**	700	200 c **700**	**1800**
DEMAND	**500**	**800**	**700**	**1400**	**3400**

(3 marks)

Row + Column – 1 = Number of Allocation

Quantity allocated		Cost/Unit ₦	Total ₦
(a) 800	*	80	= 64, 000
(b) 700	*	100	= 70, 000
(c) 700	*	200	= 140, 000
(d) 700	*	400	= 280, 000
(e) 300	*	400	= 120, 000
(f) 200	*	700	= 140, 000
Total			**814, 000**

(iii) Vogel Approximation method

	D_1	**D_2**	**D_3**	**D_4**	**Supply**	**P1**	**P2**	**P3**	**P4**
S_1	190 b **500**	300	500	100 d **200**	**700**	90	90	400	400
S_2	700	300	400 e **700**	600 f **200**	**900**	100	200	200	200
S_3	400	80 a **800**	700	200 c **1000**	**1800**	120	200	500	
DEMAND	**500**	**800**	**700**	**1400**	**3400**				
P1	210	200	100	100					
P2	210		100	100					
P3			100	100					
P4			100	500					

Row + Column – 1 = Number of Allocation

Quantity allocated	Cost/Unit ₦	Total ₦

(a) 800	*	80	= 64, 000
(b) 500	*	190	= 95, 000
(c) 1000	*	200	= 200, 000
(d) 200	*	100	= 20, 000
(e) 700	*	400	= 280, 000
(f) 200	*	600	= 120, 000
Total			**779,000**